

Project One Template

MAT350: Applied Linear Algebra

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Problem 1

Develop a system of linear equations for the network by writing an equation for each router (A, B, C, D, and E). Make sure to write your final answer as $A\mathbf{x}=\mathbf{b}$ where A is the 5×5 coefficient matrix, $\mathbf{x}$ is the 5×1 vector of unknowns, and $\mathbf{b}$ is a 5×1 vector of constants.

Solution:

Node A: $2x_1 + x_2 = 100$

Node B: $x_1 + x_2 - x_3 - x_5 = 0$

Node C: $x_1 - x_3 - x_5 = -50$

Node D: $-x_2 + x_4 + x_5 = 120$

Node E: $x_2 + x_3 - x_4 + x_5 = 0$

Linear Equation in $ax=b$ form:

$$A = \begin{bmatrix} 2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -1 & 0 & -1 \\ -1 & 0 & 1 & 0 & 1 \\ 0 & -1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 & 1 \end{bmatrix}$$

$$\mathbf{x} = [x_1; x_2; x_3; x_4; x_5]$$

$$\mathbf{b} = [100; 0; -50; 120; 0]$$

Problem 2

Use MATLAB to construct the augmented matrix $[A \ \mathbf{b}]$ and then perform row reduction using the `rref()` function. Write out your **reduced matrix and identify the free and basic variables of the system.**

Solution:

```
%code
A = [2, 1, 0, 0, 0,;
     1, 1, -1, 0, -1;
     1, 0, -1, 0, -1;
     0, -1, 0, 1, 1;
     0, 1, 1, -1, 1;]
```

```
A = 5x5
     2     1     0     0     0
```

```

1      1      -1      0      -1
1      0      -1      0      -1
0      -1      0      1      1
0      1      1      -1      1

```

```
b = [100; 0; -50; -120; 0]
```

```

b = 5x1
    100
     0
    -50
   -120
     0

```

```
x = A\b
```

```

x = 5x1
    25
    50
   270
   125
  -195

```

```
aug = [A b]
```

```

aug = 5x6
     2      1      0      0      0    100
     1      1     -1      0     -1      0
     1      0     -1      0     -1    -50
     0     -1      0      1      1   -120
     0      1      1     -1      1      0

```

```
rref(aug)
```

```

ans = 5x6
     1      0      0      0      0     25
     0      1      0      0      0     50
     0      0      1      0      0    270
     0      0      0      1      0    125
     0      0      0      0      1   -195

```

Problem 3

Use MATLAB to **compute the LU decomposition of A**, i.e., find $A = LU$. For this decomposition, find the transformed set of equations $Ly = b$, where $y = Ux$. Solve the system of equations $Ly = b$ for the unknown vector y .

Solution:

```

%code
A = [2 1 0 0 0; 1 1 -1 0 -1; -1 0 1 0 1; 0 -1 0 1 1; 0 1 1 -1 1]

```

```

A = 5x5
     2      1      0      0      0
     1      1     -1      0     -1
    -1      0      1      0      1
     0     -1      0      1      1
     0      1      1     -1      1

```

```
b = [100; 0; 50; 120; 0]
```

```
b = 5x1
    100
     0
     50
    120
     0
```

```
[L U] = lu(A)
```

```
L = 5x5
    1.0000     0     0     0     0
    0.5000   -0.5000   -1.0000    1.0000     0
   -0.5000   -0.5000    1.0000     0     0
     0     1.0000     0     0     0
     0   -1.0000    1.0000   -0.5000    1.0000

U = 5x5
    2.0000    1.0000     0     0     0
     0   -1.0000     0    1.0000    1.0000
     0     0    1.0000    0.5000    1.5000
     0     0     0    1.0000    1.0000
     0     0     0     0    1.0000
```

```
y = linsolve(L, b)
```

```
y = 5x1
    100
    120
    160
    170
     45
```

Problem 4

Use MATLAB to **compute the inverse** of U using the `inv()` function.

Solution:

```
%code
inv(U)
```

```
ans = 5x5
    0.5000    0.5000     0   -0.5000     0
     0   -1.0000     0    1.0000     0
     0     0    1.0000   -0.5000   -1.0000
     0     0     0    1.0000   -1.0000
     0     0     0     0    1.0000
```

Problem 5

Compute the solution to the original system of equations by transforming **y** into **x**, i.e., compute **x** = `inv(U)y`.

Solution:

```
%code
```

```
x = inv(U)*y
```

```
x = 5x1
    25
    50
    30
   125
    45
```

Problem 6

Check your answer for x_1 using Cramer's Rule. Use MATLAB to compute the required determinants using the `det()` function.

Solution:

```
%code
```

```
A1 = A
```

```
A1 = 5x5
```

```
    2    1    0    0    0
    1    1   -1    0   -1
   -1    0    1    0    1
    0   -1    0    1    1
    0    1    1   -1    1
```

```
A2 = A
```

```
A2 = 5x5
```

```
    2    1    0    0    0
    1    1   -1    0   -1
   -1    0    1    0    1
    0   -1    0    1    1
    0    1    1   -1    1
```

```
A3 = A
```

```
A3 = 5x5
```

```
    2    1    0    0    0
    1    1   -1    0   -1
   -1    0    1    0    1
    0   -1    0    1    1
    0    1    1   -1    1
```

```
A4 = A
```

```
A4 = 5x5
```

```
    2    1    0    0    0
    1    1   -1    0   -1
   -1    0    1    0    1
    0   -1    0    1    1
    0    1    1   -1    1
```

```
A5 = A
```

```
A5 = 5x5
```

```
    2    1    0    0    0
```

1	1	-1	0	-1
-1	0	1	0	1
0	-1	0	1	1
0	1	1	-1	1

A1(:,1) = b

A1 = 5x5

100	1	0	0	0
0	1	-1	0	-1
50	0	1	0	1
120	-1	0	1	1
0	1	1	-1	1

A2(:,2) = b

A2 = 5x5

2	100	0	0	0
1	0	-1	0	-1
-1	50	1	0	1
0	120	0	1	1
0	0	1	-1	1

A3(:,3) = b

A3 = 5x5

2	1	100	0	0
1	1	0	0	-1
-1	0	50	0	1
0	-1	120	1	1
0	1	0	-1	1

A4(:,4) = b

A4 = 5x5

2	1	0	100	0
1	1	-1	0	-1
-1	0	1	50	1
0	-1	0	120	1
0	1	1	0	1

A5(:,5) = b

A5 = 5x5

2	1	0	0	100
1	1	-1	0	0
-1	0	1	0	50
0	-1	0	1	120
0	1	1	-1	0

x1 = det(A1)/det(A)

x1 =
25.0000

x2 = det(A2)/det(A)

x2 =
50

x3 = det(A3)/det(A)

x3 =
30.0000

x4 = det(A4)/det(A)

x4 =
125.0000

x5 = det(A5)/det(A)

x5 =
45

Problem 7

The Project One Table Template, provided in the Project One Supporting Materials section in Brightspace, shows the recommended throughput capacity of each link in the network. Put your solution for the system of equations in the third column so it can be easily compared to the maximum capacity in the second column. In the fourth column of the table, provide recommendations for how the network should be modified based on your network throughput analysis findings. The modification options can be No Change, Remove Link, or Upgrade Link. In the final column, explain how you arrived at your recommendation.

Solution:

*Fill out the table in the original project document and export your table as an image. Then, use the **Insert** tab in the MATLAB editor to insert your table as an image.*

MAT 350 Project One Table Template

Complete this template by replacing the bracketed text with the relevant information.

Network Link	Recommended Capacity (Mbps)	Solution	Recommendation	Explanation
x ₁	60	25	No Change	Solution is within link capacity
x ₂	50	50	No Change	Solution is within link capacity
x ₃	100	30	No Change	Solution is within link capacity
x ₄	100	125	Upgrade	Recommended link capacity is less than the because of this it needs to be upgraded.
x ₅	50	45	No Change	Solution is within link capacity